

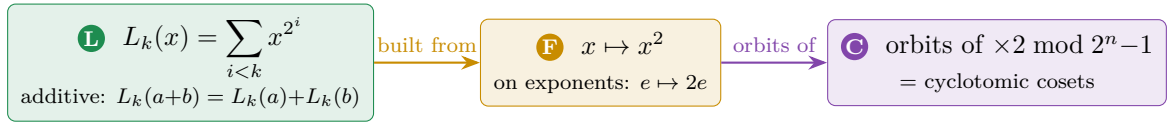
Dobbertin (1999): one gadget, many theorems

A picture book of Kasami power functions, permutation polynomials and cyclic difference sets

Visual summary — Dobbertin1999MVP (Lean / Mathlib)

Thesis. Every result of the paper is the *same small gadget* seen from a new angle. The gadget has three faces: **L** the **linearized trace** $L_k(x) = \sum_{i < k} x^{2^i}$ (a truncated, additive polynomial); **F** **Frobenius** $x \mapsto x^2$ (“freshman’s dream”, doubling on exponents); **C** **cyclotomic-coset** book-keeping (orbits of doubling mod 2^n-1). The permutation criterion, APN, the MCM bridge, the explicit inverse, and the difference sets are all this one gadget, propagated. Boxes marked ✓ are machine-checked in Lean.

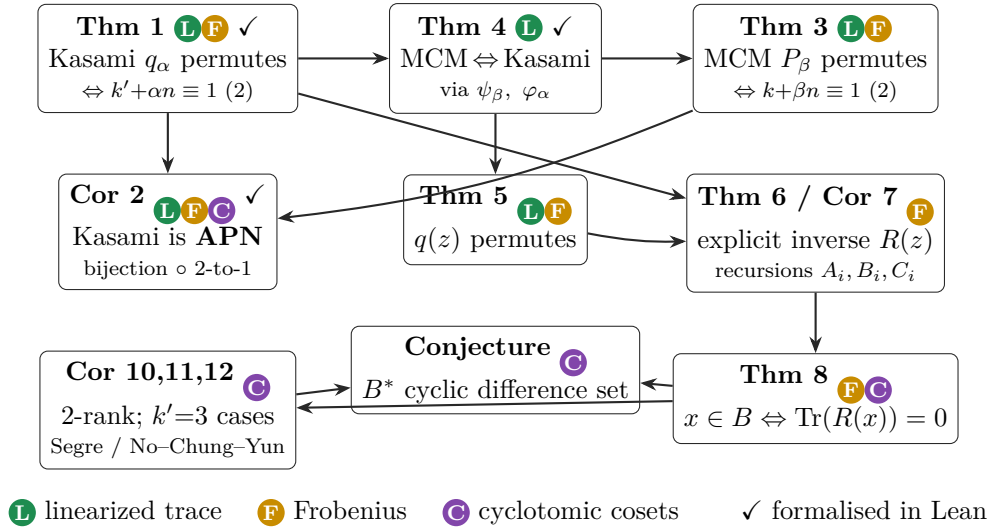
0. The gadget



Telescoping identity that glues **L** to **F**: $L_k(x^2 + x) = x^{2^k} + x$ (`truncTrace_artin_schreier`, generalised in `Pearl.lean`) ✓

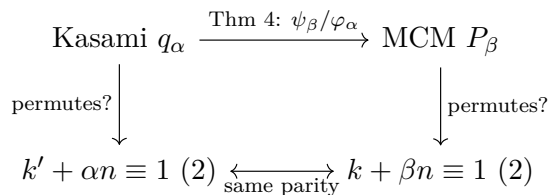
1. The whole paper on one page (dependency map)

Arrows = “used to prove”. Each node carries the faces of the gadget it leans on.



2. Permutation criterion (Thm 1 = Thm 3): a parity square

Both criteria have the *same shape*: “ q_α (resp. P_β) permutes $\iff$ a single parity is odd”. The operator on the left of the linearized equation is the gadget **L**.



3. Theorem 4: the bridge is an isomorphism of self-maps

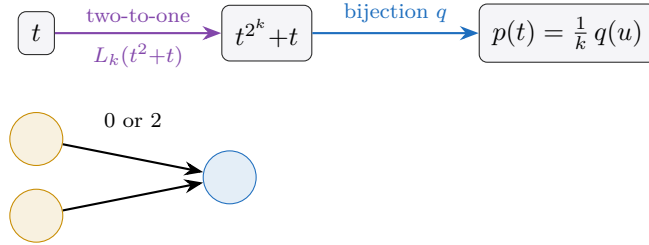
$\psi_\beta(x) = \sum_{i < k} x^{2^i} + \beta \text{Tr}(x)$ (that is **L**, plus a trace) is a *bijective intertwiner*: it conjugates the Kasami power map into the MCM power map. In the category **Endo** of sets-with-a-self-map this is one iso.

$$\begin{array}{ccc}
 L & \xrightarrow{\psi_\beta} & L \\
 \varphi_\alpha \swarrow & & \nwarrow \varphi_\alpha \\
 L & \xrightarrow{\psi_\beta} & L \\
 \varphi_\alpha \swarrow & & \nwarrow \varphi_\alpha
 \end{array}
 \quad
 \begin{array}{c}
 q_\alpha \downarrow \\
 P_\beta \downarrow
 \end{array}
 \quad
 P_\beta = \psi_\beta \circ q_\alpha \circ \psi_\beta^{-1}$$

Lean: `mcm_permutation, mcm_permutation_ktransfer (✓)`; category `home Endo/isoOfEquiv (ShiftConjugacy.lean, ✓)`.

4. Corollary 2: APN = (bijection) $\circ$ (two-to-one)

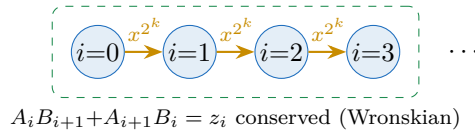
The Kasami derivative factors through the gadget. The right factor is the *Artin-Schreier* map $t \mapsto t^{2^k} + t = L_k(t^2 + t)$ — exactly two-to-one because $\gcd(k, n) = 1$ (**C**). Bijection $\circ$ 2-to-1 $\Rightarrow$ every fibre has 0 or 2 points $\Rightarrow$ APN.



Key identity (the concrete Gold $\leftrightarrow$ Kasami bridge, $\checkmark$): $((x+1)^d + x^d + 1)(x^2 + x)^{2^k} = (x^{2^k} + x)^{2^k + 1}$
 Lean: `kasami_key_identity, gold_permutation, kasami_is_apn, kasami_is_apn_solution_count`.

5. Theorem 6 / Cor 7: the same shift, iterated $\Rightarrow$ inverse

Feeding the linearized equation through Frobenius **F** repeatedly spawns one relation per level i (eq. (10)); the three families A_i, B_i, C_i obey *one* two-step recursion. Their 2×2 Wronskian cross-invariants (eqs. (11)–(13)) are conserved, which collapses the would-be cubic to a line: the unique root is the explicit inverse R .

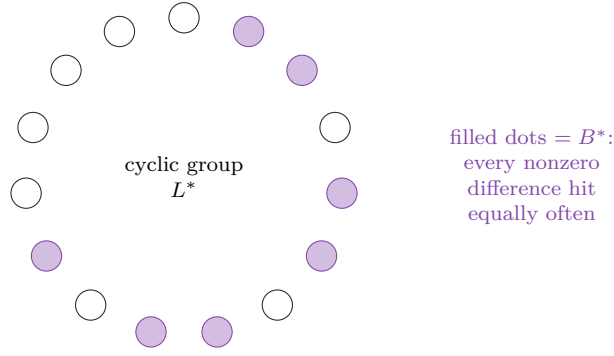


$$\text{cubic } x^3 + A_2 x^2 + A_1 x + A_0 \xrightarrow{A_1=A_0=0} \text{line } x = R(y) \xrightarrow{q^{-1}(1/y)=R(y)} \text{explicit inverse}$$

Conserved cross-invariant formalised: `Theorem6.cross_invariant, Theorem6.key_identity (✓, char. 2 cancellation)`.

6. Theorem 8 + Conjecture: the coset combinatorics of B

R from Thm 6 defines the set $B = \{x : \text{Tr}(R(x)) = 0\}$ (Thm 8, a trace test **F**). The Section-4 conjecture is pure **C**: B^* is a difference set in the cyclic group L^* , and the $k \leftrightarrow n-k$ / $(k, k') \leftrightarrow (n-k, n-k')$ symmetries are cyclotomic shifts.



Conjecture (up to cyclotomic equivalence): only $d = 2^k + 1$ (Gold) and $d = 2^{2k} - 2^k + 1$ (Kasami) yield this B ; contains Segre / Maschietti (1998) and No-Chung-Yun (1998) / Dillon $k'=3$ cases.

7. Why it is “one pattern”: the census as a table

Result	Gadget face used	Lean
Thm 1 — Kasami permutation criterion	L operator, F shift	<code>mcm_permutation_ktransfer</code> ✓
Cor 2 — Kasami APN	L ◦ F telescope, C 2-to-1	<code>kasami_is_apn</code> ✓
Thm 3 — MCM permutation criterion	L operator (dual)	via Thm 4
Thm 4 — MCM ⇔ Kasami (ψ_β)	L = intertwiner	<code>mcm_permutation</code> ✓
Thm 5 — q is a permutation	L , F	(implied)
Thm 6 / Cor 7 — explicit inverse R	F iterated, Wronskian	<code>Theorem6.*</code> ✓
Thm 8 — difference set B	F trace test, C	—
Cor 10/11/12 — 2-rank, $k'=3$	C coset counting	—
Conjecture — B^* difference set	C cyclotomic equivalence	—

One sentence. The linearized trace **L** and Frobenius **F** give the permutation criterion; telescoping **L**◦**F** plus a two-to-one Artin–Schreier map (**C**) gives APN; the bijective intertwiner **L** gives the MCM bridge; iterating **F** with conserved Wronskians gives the inverse; and coset bookkeeping **C** carries it all into the difference sets — one gadget, propagated from the smallest identity to the global conjecture.

8. Formal status (what is machine-checked)

✓ Artin–Schreier telescope <code>truncTrace_artin_schreier, Pearl.lean</code>	✓ MCM bridge ψ_β <code>mcm_permutation(_ktransfer)</code>	✓ APN <code>kasami_is_apn(_solution_count)</code>
✓ key identity <code>kasami_key_identity</code>	✓ Wronskian invariant <code>Theorem6.cross_invariant</code>	✓ conjugacy category <code>Endo, ShiftConjugacy.lean</code>